

NAME: Mary Nyambura Gitau
GMAIL:marynyambura657@gmail.com
ADMISSION NUMBER: CS-SA12-26067

WIFI HACKING 101 WITH TRY HACK ME

1.1 Introduction

In this module, I will explore the fundamentals of Wi-Fi security using the TryHackMe platform. The focus of the module is to understand how attackers target wireless networks that use weaker security configurations such as WPA. I will learn how wireless communication works, how attackers can intercept or analyze network traffic, and how tools such as Wireshark can be used to capture and inspect packets moving across a network. By performing these practical exercises, I will gain insight into common wireless network vulnerabilities and how they can be exploited. This knowledge will also help me understand the importance of securing wireless networks and implementing stronger protection mechanisms against such attacks.

1.2 The basic : an intro to WPA

- SSID: The network "name" that you see when you try and connect
- ESSID: An SSID that *may* apply to multiple access points, eg a company office, normally forming a bigger network. For Aircrack they normally refer to the network you're attacking.
- BSSID: An access point MAC (hardware) address
- WPA2-PSK: Wifi networks that you connect to by providing a pre-shared password that's the same for everyone
- WPA2-EAP: Wifi networks that you authenticate to by providing a username and password, which is sent to a RADIUS server.
- RADIUS: A server for authenticating clients, not just for wifi.

a) What type of attack on the encryption can you perform on WPA(2) personal?

Answer: **Brute force**

b) Can this method be used to attack WPA2-EAP handshakes? (Yea/Nay)

Answer: **Nay**

c) What is the three-letter abbreviation for the pre-shared key used in Wi-Fi security?

Answer: **PSK**

d)What's the minimum length of a WPA2 Personal password?

Answer: **8**

Below is a screenshot with the evidence of my answers

The screenshot shows a quiz interface with the title "Answer the questions below". It contains four questions, each with a text input field for the answer and a green button labeled "✓ Correct Answer".

- Question 1: "What type of attack on the encryption can you perform on WPA(2) personal?"
Answer: "brute force"
- Question 2: "Can this method be used to attack WPA2-EAP handshakes? (Yea/Nay)"
Answer: "Nay"
- Question 3: "What is the three-letter abbreviation for the pre-shared key used in Wi-Fi security?"
Answer: "PSK"
- Question 4: "What's the minimum length of a WPA2 Personal password?"
Answer: "8"

1.3 You are being watched -capturing packets

a)How do you put the interface “wlan0” into monitor mode with Aircrack tools? (Full command)

Answer: **airmon-ng start wlan0**

b)What is the new interface name likely to be after you enable monitor mode?

Answer: **wlan0mon**

c)What do you do if other processes are currently trying to use that network adapter?

Answer: **airmon-ng check kill**

d)What tool from the aircrack-ng suite is used to create a capture?

Answer: **airodump-ng**

e)What flag do you use to set the BSSID to monitor?

Answer: **--bssid**

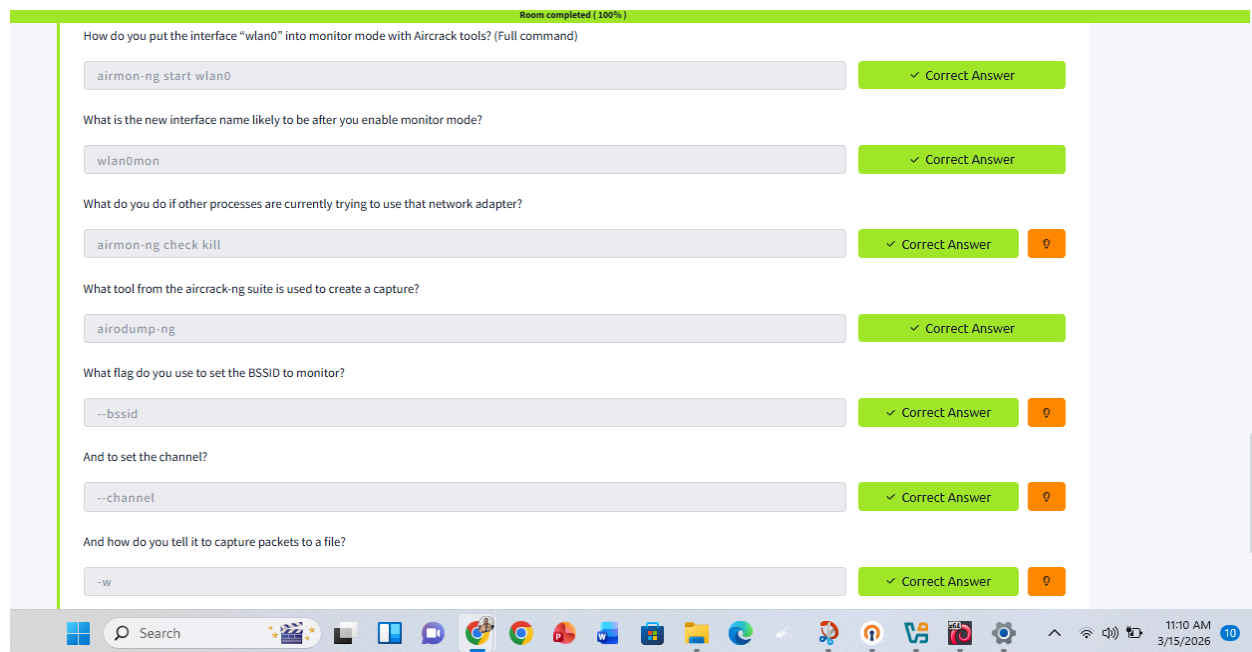
f)And to set the channel?

Answer: **--channel**

g)And how do you tell it to capture packets to a file?

Answer: **-w**

Below is a screenshot of my answers



1.4 Aircrack-ng :lets get cracking

a)What flag do we use to specify a BSSID to attack?

Answer: **-b**

b)What flag do we use to specify a wordlist?

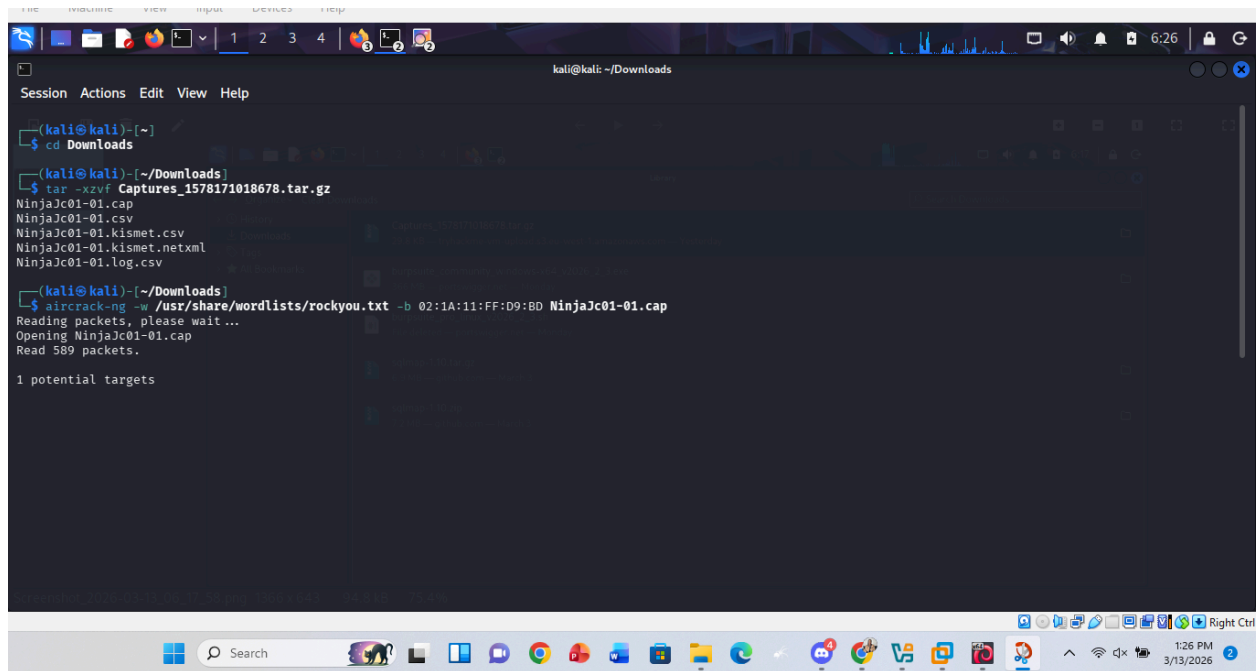
Answer: **-w**

c)How do we create a HCCAPX in order to use hashcat to crack the password?

Answer: **-j**

d)Using the rockyou wordlist, crack the password in the attached capture. What's the password?

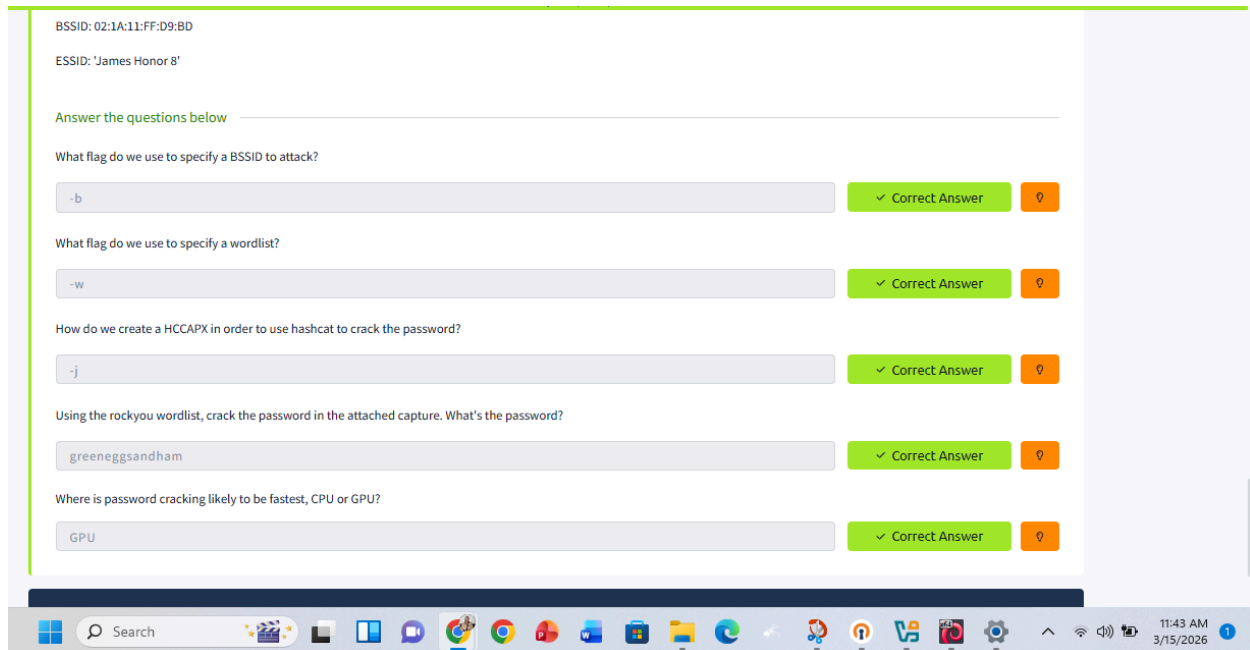
Answer: **greeneggsandham**



```
(kali@kali)-[~]
└─$ cd Downloads
(kali@kali)-[~/Downloads]
└─$ tar -xvzf Captures_1578171018678.tar.gz
NinjaJc01-01.cap
NinjaJc01-01.csv
NinjaJc01-01.kismet.csv
NinjaJc01-01.kismet.netxml
NinjaJc01-01.log.csv
(kali@kali)-[~/Downloads]
└─$ aircrack-ng -w /usr/share/wordlists/rockyou.txt -b 02:1A:11:FF:D9:BD NinjaJc01-01.cap
Reading packets, please wait...
Opening NinjaJc01-01.cap
Read 589 packets.

1 potential targets
```

Below is the screenshot of my answers.



Conclusion and Summary

In this module, I learned the basics of Wi-Fi security and how attackers can target wireless networks with weaker encryption such as WPA. I also learned how to use tools like Wireshark to capture and analyze network packets in order to understand how data travels across a network. Through the practical exercises on TryHackMe, I gained a better understanding of how wireless attacks work and why it is important to secure networks properly. This module helped me build foundational knowledge in wireless network security and traffic analysis.

Below is a screenshot of the completion of the module.

<https://tryhackme.com/p/marynyambura657>

Share your win with your peers

Your progress can inspire others in your community.



Wifi Hacking 101 completed!

marynyambura657 completed another room on their cyber security journey.

Completed tasks	Points earned	Streak
 3	 128	 2

Share on social media

Share your win before starting the next room

[in LinkedIn](#)[WhatsApp](#)[Telegram](#)[X Twitter / X](#)[Facebook](#)[Reddit](#)

[Go to dashboard →](#)

